

BREAK INTO AEROSPACE

Earth Observation Engineering

Subdomain Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

Earth Observation is one of the fastest-restructuring sectors in aerospace engineering. What was once a field defined by a handful of government satellite programmes is now a sprawling commercial industry operating constellations of dozens to hundreds of spacecraft, producing petabytes of imagery and data annually, and serving customers ranging from agricultural insurers to defence intelligence agencies to humanitarian response organisations. The technical demands of EO span optical physics and detector science, radar signal processing, orbit mechanics, radiometric and geometric calibration, atmospheric physics, machine learning, and large-scale data engineering — all in the service of extracting reliable, quantitative information about the Earth's surface and atmosphere from instruments operating 400 to 800 kilometres above it.

Engineers in demand across an exceptionally varied employer landscape: legacy primes with satellite manufacturing heritage (Airbus D&S, Thales Alenia Space, OHB, Ball Aerospace, Northrop Grumman), new-space commercial operators (Planet Labs, Maxar, ICEYE, Capella Space, Satellogic, BlackSky, SatVu, Pixxel, Wyvern, Umbra), specialist sensor manufacturers (Teledyne DALSA, e2v, FLIR, Xenics), ground segment and data processing companies (Sinergise, ENVEO, DHI GRAS), and government agencies running the world's largest EO programmes (ESA Copernicus, NASA Landsat/PACE/NISAR, JAXA ALOS, DLR TerraSAR-X).

PART I · WHAT THE EO FIELD ACTUALLY LOOKS FOR

By Level

- **Intern / Graduate** — At entry level, EO employers are not primarily screening for satellite or remote sensing expertise — they are screening for physical intuition, mathematical rigour, and the ability to think quantitatively about signal and noise. The questions that distinguish strong from weak graduate candidates are not “what is a SAR satellite?” but “if I double my telescope aperture, what happens to my spatial resolution and my signal-to-noise ratio, and do these always move together?” or “what does the diffraction limit mean for a pushbroom imager trying to achieve 30 cm ground sample distance from 500 km orbit?”

Graduate interviews at larger employers — Airbus D&S, Thales Alenia Space, OHB, Ball Aerospace, Surrey Satellite Technology — involve a technical screen covering optics fundamentals, signal processing, or programming, followed by interviews mixing domain conceptual questions with behavioural probing. At smaller new-space operators — Planet, ICEYE, Capella, Pixxel — the screen is often a take-home exercise or data challenge involving real satellite imagery or radiometric data. The data challenge is the distinguishing feature: new-space EO companies assess whether you can *do something useful with an image*, not just explain how one is formed.

- **Mid-Level (Engineer, Senior Engineer, ~3–8 years)** — At mid-level, EO interviews become domain-specific and problem-oriented. Questions shift from “explain atmospheric correction” to “your Level 2 surface reflectance product shows a 5% bias over bright desert targets compared to your vicarious calibration reference — walk me through your diagnostic approach” or “we are designing a hyperspectral instrument for a 500 km SSO orbit and need 10 nm spectral resolution with SNR greater than 300:1 for vegetation targets — what detector technology and integration time are you designing toward, and what

spatial resolution does that force?”

The specific context driving mid-level hiring in 2025–2026: European defence spending post-2022 is creating demand for sovereign EO intelligence capability (SAR and optical tasking for tactical reconnaissance, persistent monitoring, damage assessment); the Copernicus expansion (CO2M, CHIME, CIMR, LSTM) is creating demand for calibration, ground segment, and data processing engineers; the precision agriculture market is driving demand for frequent multispectral revisit and ML applied to canopy reflectance; and the maritime domain awareness market is creating demand for combined AIS/SAR/optical fusion capability.

- **Senior (Principal Engineer, Technical Lead, Fellow)** — At senior level, EO employers look for engineers who can define the technical architecture of a complete EO system — from instrument specification through calibration approach through ground processing chain — and who can make and defend the trade-offs that determine what users get at the end. The senior interview is typically a peer discussion with the technical leadership: you may be asked to assess a proposed instrument concept, identify the failure modes in a calibration approach, or critique an organisation’s current image quality pipeline.

Senior roles at mission integrators (Airbus, Thales Alenia, OHB), space agencies (ESA, NASA JPL, DLR), and established commercial operators (Maxar, Planet, Satellogic) require engineers who have taken a real instrument or processing chain from specification through on-orbit commissioning and demonstrated calibration stability over time. The on-orbit commissioning experience is specifically valued — the first weeks after launch, when unexpected thermal, orbital, or electrical environments reveal discrepancies from ground calibration, are a crucible that cannot be simulated.

Hands-On Experience and Domain Knowledge Treated as Strong Signals

Domain	Why It Signals Depth
Radiometric Calibration	The quantitative value of EO data depends entirely on how accurately digital numbers from a detector array can be converted to physical units of radiance or reflectance. Engineers who have designed, implemented, or validated a radiometric calibration chain — not just applied a coefficient — understand where the errors live and how they propagate into downstream products.
Geometric Calibration and Orthorectification	Engineers who have implemented or validated the geometric calibration and orthorectification chain — specifically the bundle adjustment of ground control points and the propagation of residual geolocation error into product specifications — demonstrate depth that distinguishes imagery providers from imagery users.
SAR Processing	Engineers who have implemented any part of the SAR processing chain — even a range-Doppler algorithm on simulated data — distinguish themselves. Engineers who understand coherence in repeat-pass InSAR, and what temporal and spatial decorrelation do to it, are specifically valued in the growing SAR analytics market.
Atmospheric Correction	Engineers familiar with radiative transfer codes (MODTRAN, 6S, libRadtran), operational atmospheric correction processors (Sen2Cor, ACOLITE, LaSRC, MAJA), and the uncertainty sources in each are demonstrating the depth that quantitative remote sensing requires.
Hyperspectral and Thermal Sensor Design	The commercial hyperspectral market is in early rapid growth (Pixxel, Wyvern, HyperSat, Copernicus CHIME), creating specific demand for engineers with experience in InGaAs, HgCdTe, or microbolometer detector technology and the processing algorithms specific to the physics of each band.
Machine Learning Applied to EO	Engineers who understand the domain-specific challenges of satellite imagery — varying illumination, sparse and geographically biased ground truth, temporal and spectral distribution shift — and can design training and evaluation frameworks that account for them are more valuable than engineers who apply standard computer vision without domain understanding.
Ground Segment and Data Pipeline Engineering	Engineers with production experience in distributed satellite data processing — Apache Spark or Dask for large-raster processing, Zarr or NetCDF for multidimensional data, STAC for metadata — are increasingly valued as commercial EO transitions from a science to an operational data service.
Image Quality Specification and Measurement	Engineers who can specify GSD, MTF, SNR, NEdL/NEdT, and Dynamic Range for a mission, derive them from instrument parameters, measure them from on-orbit data, and interpret them in terms of application capability are demonstrating the systems engineering depth that links payload design to end-user requirements.

How EO Interviews Differ from Generic Aerospace or Software Roles

- **The physics of electromagnetic radiation is unavoidable.** A mid-level EO engineer who cannot explain why a vegetation canopy reflects strongly in the near-infrared (the “red edge” effect driven by mesophyll cell scattering), why SAR backscatter from calm water is low (specular reflection away from the sensor), or what the diffraction limit implies for achievable GSD at a given aperture and wavelength has a gap that is immediately apparent in a technical interview. The physics is not background knowledge — it is the source of every constraint in the system.
- **The distinction between sensor modalities is deep, not superficial.** EO encompasses optical (panchromatic, multispectral, hyperspectral), thermal infrared, synthetic aperture radar, and radio occultation sensors. These are not variations on a theme — they are fundamentally different physical measurement systems with different strengths, limitations, calibration challenges, and processing requirements. Interviewers who work in one modality daily will probe its specifics without adjustment.
- **Working with real EO data is expected, not optional.** Sentinel-2, Landsat-9, Sentinel-1, SRTM, ICESat-2, and many other datasets are freely available through ESA’s Copernicus Data Space, USGS Earth Explorer, and the AWS/Google Cloud open data programmes. Candidates above intern level who have not processed real EO data — not just looked at images in a GIS but processed them programmatically — are demonstrably less prepared than those who have.
- **Calibration is the engineering discipline that determines whether EO data has any value.** An uncalibrated image is a picture. A calibrated, validated, traceable image is a measurement. Interviewers at instrument manufacturers, mission operators, and data processing companies will probe calibration understanding at depth. “How would you detect an on-orbit radiometric drift in your instrument?” is a standard mid-level question.
- **New-space and legacy-space cultures have genuinely different interview formats.** A Planet Labs interview involves a take-home data challenge before any phone screen. An Airbus Defence & Space interview involves structured technical panels over two rounds. ESA interviews include a written test and panel, often months apart. SSTL interviews quickly and informally relative to larger primes.
- **Security clearance is mission-critical for defence EO roles.** Maxar Intelligence, Airbus Intelligence, BlackSky, and the defence EO divisions of Northrop Grumman, L3Harris, and BAE Systems work with classified imagery. US clearance (typically TS/SCI with CI polygraph for NRO-adjacent work) is required for most US defence EO roles. UK DV clearance is required for equivalent UK roles.

PART II · 20 SPECIFIC PREPARATION TASKS

Intern / Graduate

1. Process a Complete Sentinel-2 Scene from Download to Surface Reflectance — Programmatically, Without a GUI 6–10 h

Skill: End-to-end EO data processing competence as the field’s baseline practical signal

Objective: Sentinel-2 is the world’s most widely used open EO dataset — 13 spectral bands, 10m/20m/60m resolution, 5-day global revisit, free access through the Copernicus Data Space. “Programmatically, without a GUI” is the key qualifier: downloading a scene and opening it in QGIS demonstrates nothing about ability to contribute to a data processing pipeline. Writing Python that downloads the scene using the Copernicus API, reads the Level 1C top-of-atmosphere reflectance bands using rasterio or GDAL, applies Sen2Cor or py6S atmospheric correction to produce surface reflectance, calculates NDVI, applies a basic cloud mask using the SCL band, and saves a georeferenced GeoTIFF with correct CRS — this demonstrates the practical literacy that distinguishes em-

ployable candidates from informed bystanders.

6–10 h

Done Signal: You have a working Python script — ideally on GitHub — that downloads a Sentinel-2 L1C scene, applies atmospheric correction, calculates at least one vegetation or water index, applies cloud masking, and saves a georeferenced multi-band output. You can explain every processing step, the physical meaning of each band used, and the sources of uncertainty in the surface reflectance estimate.

2. Understand the Electromagnetic Spectrum as an EO Engineering Constraint — What Each Region Can and Cannot See, and Why

3–4 h

Skill: Spectral physics as the foundational constraint of every EO instrument design decision

Objective: EO instruments are designed to observe specific spectral regions because different regions interact with the Earth's surface and atmosphere in physically different ways. Visible and near-infrared radiation reflects from the surface and is scattered by aerosols. Shortwave infrared penetrates thin clouds and is sensitive to soil moisture and snow. Thermal infrared is emitted by the surface according to its temperature and emissivity. Microwave radar penetrates cloud, rain, and vegetation canopy and is sensitive to surface roughness and dielectric properties. Candidates who do not know which atmospheric windows exist, why SAR can image through cloud when optical cannot, or what the red edge transition in vegetation spectra means will signal immediately that their engagement with EO is superficial.

Done Signal: For each major EO spectral region — visible (400–700 nm), near-infrared (700–1300 nm), shortwave infrared (1300–2500 nm), thermal infrared (8–12 μm), and microwave (1–100 cm) — you can state: what the dominant surface interaction is, what the dominant atmospheric effects are, what the primary EO application is, and one instrument or mission that operates in this band. You can explain why different spectral bands respond differently to cloud cover.

3. Understand Image Quality Metrics — GSD, MTF, SNR, NEdL — and How They Connect Instrument Parameters to Application Capability

3–4 h

Skill: The system-level language that links payload design to end-user requirements

Objective: Every EO instrument specification is written in terms of image quality metrics: “GSD 3 m at nadir, SNR > 200:1 for typical vegetation target at 45° solar zenith, MTF > 0.2 at Nyquist.” These are derived from application requirements and constrained by physics. Candidates who cannot explain what MTF measures, why it matters for target recognition, and what instrument parameters drive it are demonstrating they have read about EO without understanding the engineering vocabulary that connects design to performance.

Done Signal: You can explain what GSD is and how it relates to aperture focal length and detector pixel size. You can explain what MTF measures — the contrast at a given spatial frequency — and why a camera with high GSD but poor MTF may perform worse on target detection than a camera with lower GSD but excellent MTF. You can explain what SNR means for a pushbroom sensor, what NEdL represents, and what integration time trade-offs it implies. You can describe one method for measuring MTF from on-orbit imagery.

4. Prepare Behavioural Stories That Connect Technical Work to Data Quality or Application Impact

2–3 h

Skill: STAR-format stories with the quality-and-outcome emphasis EO employers probe

Objective: EO interviewers specifically probe attention to data quality, validation rigour, and the consequences of errors that propagate silently through a processing chain. A candidate who has

2–3 h

detected a silent error in their own analysis — a coordinate reference system mismatch, a radiometric scaling error, a cloud mask failure that produced a biased vegetation index — and corrected it before it affected downstream outputs is demonstrating the quality discipline EO data products require. Stories that end with “the analysis looked right” are the weakest. Stories that end with “we validated against an independent reference and found a 3% bias that we traced to a misconfigured atmospheric correction parameter” are the strongest.

Done Signal: You have four STAR stories covering: a data quality issue you detected and diagnosed, a time you worked across disciplinary boundaries (science and engineering, processing and validation), a technical mistake in your own analysis that you caught and corrected before it reached a stakeholder, and a collaboration that produced a result neither party could have achieved alone. At least two stories involve real EO data.

3–4 h

5. Understand GNSS-Based Georeferencing and Why Geolocation Accuracy Matters for EO Products

Skill: The geometric foundation of all EO product value

Objective: An EO image that cannot be accurately located on the Earth’s surface is not useful for most applications. Geolocation accuracy depends on instrument pointing knowledge (star tracker accuracy, gyroscope noise), orbital position accuracy, instrument geometric calibration, and terrain correction. Candidates who understand that a 1-arcsecond pointing error at 500 km orbit produces a 2.4 m geolocation error, or what the trade-off is between using ground control points and relying purely on sensor geometry models, are demonstrating the physical reasoning that instrument engineers and product managers rely on.

Done Signal: You can explain what a rational polynomial coefficient (RPC) model is and why it is the standard geolocation model for high-resolution commercial imagery. You can describe what ground control points contribute to geolocation accuracy and what the achievable accuracy is with and without them for a typical commercial 0.5 m GSD satellite. You can explain what terrain displacement means in the context of a satellite with 30° off-nadir viewing angle and a 1000 m hill.

5–8 h

6. Get Hands-On with SAR Data — Download a Sentinel-1 GRD Product, Process It, and Understand What You Are Looking At

Skill: Practical SAR literacy in an instrument modality that is increasingly commercially dominant

Objective: SAR imagery is growing faster than any other EO modality commercially — ICEYE, Capella Space, Umbra, Synspective, and Satellogic SAR are all operational or launching; the NRO’s future architecture heavily favours SAR for its all-weather, day/night capability; and Sentinel-1 provides free SAR data globally. Yet SAR is substantially less intuitive than optical imagery and requires different interpretation skills: backscatter intensity depends on surface roughness, dielectric properties, and incidence angle in ways that are non-obvious without domain knowledge.

Done Signal: You have downloaded a Sentinel-1 IW GRD product from the Copernicus Data Space, processed it using either snappy (Python bindings for ESA SNAP), pyroSAR, or sarpy, applied at least one processing step beyond the default (speckle filtering, terrain correction using an external DEM, or polarimetric decomposition using a dual-pol dataset), and can explain why the backscatter values differ between two land cover types visible in your processed scene.

Mid-Level (~3–8 years)

All graduate tasks still apply. Behavioural stories must come from professional contexts involving real instruments, real data pipelines, or real products delivered to real users.

7. Build a Quantitative Understanding of the EO Instrument Link Budget — from Solar Irradiance to Detector Digital Number 4–6 h

Skill: The radiometric chain as an engineering calculation, not a conceptual description

Objective: Every EO instrument design starts with a link budget: given the solar irradiance at the top of atmosphere, the target reflectance, the atmospheric transmission, the telescope aperture, the optical transmission, the detector quantum efficiency, the integration time, and the detector noise floor — what is the signal-to-noise ratio for the specified target? Getting this calculation right determines whether an instrument concept is feasible before any hardware is built. Getting it wrong is discovered during the first calibration campaign, when the instrument either saturates on bright targets or produces unusable data on dark ones.

Done Signal: You can work through a simplified radiometric link budget for a visible-band push-broom sensor: starting from solar irradiance (approximately $1700 \text{ W/m}^2/\mu\text{m}$ at 550 nm), applying target reflectance, atmospheric transmission, aperture area, pixel solid angle, optical transmission, detector QE, and integration time to arrive at a photon count — then computing the shot-noise limited SNR. You can identify which parameters most strongly drive SNR and what the trade-offs are between integration time, aperture, GSD, and spectral bandwidth. You can do this in under 10 minutes with a calculator.

8. Understand SAR Processing in Depth — Range-Doppler Algorithm, InSAR Coherence, and What Limits Resolution 6–8 h

Skill: The technical core of the most complex EO instrument modality

Objective: SAR processing is mathematically demanding and physically subtle. The range-Doppler algorithm applies a matched filter in the range dimension, exploits the synthetic aperture formed by the satellite's along-track motion to achieve azimuth resolution, and corrects for the range migration that occurs as each target moves through different range cells during the integration time. Understanding why the azimuth resolution of a SAR equals half the antenna length — a result that is counterintuitive until the synthetic aperture geometry is understood — and what limits range resolution (chirp bandwidth) distinguishes a SAR engineer from a SAR user. InSAR enables millimetre-scale surface deformation measurement; understanding coherence and temporal decorrelation is specifically required at companies working on deformation monitoring, ice sheet dynamics, or infrastructure monitoring.

Done Signal: You can explain why SAR azimuth resolution is independent of range and equals half the real aperture length. You can describe what range migration is and why it must be corrected during focusing. You can explain what InSAR coherence measures, what causes temporal decorrelation, and why C-band (Sentinel-1) decorrelates faster than L-band (ALOS-2/NISAR) over vegetated areas. You have processed at least one InSAR pair — even from publicly available Sentinel-1 SLC data — and interpreted the resulting interferogram.

9. Prepare a Technically Specific Story About Discovering and Correcting a Calibration Anomaly in On-Orbit or Processed EO Data 2 h

Skill: Calibration diagnosis as applied engineering rigour

Objective: On-orbit calibration anomalies are the defining technical challenge of EO instrument operations. Detector aging changes responsivity. Contamination of optical surfaces reduces transmission. Thermal cycling causes focal plane registration shifts. Stray light contaminates dark-target measurements. Each anomaly manifests as a specific pattern in calibration data — a monotonic trend in vicarious calibration residuals, a spatial pattern in flat-field coefficients, a spectral artifact in onboard calibration lamp spectra — and each requires a specific diagnostic approach and correction. Engineers who have diagnosed a real calibration anomaly and can describe the data they examined, the

hypothesis they formed, the test they designed to confirm it, and the correction they implemented are demonstrating applied calibration engineering that textbook knowledge cannot substitute. 2 h

Done Signal: You have a story about a radiometric or geometric anomaly in EO data that you diagnosed and corrected. The story names the anomaly (a specific striping pattern, a spectral artifact, a geolocation offset, a temporal drift), the data you examined to characterise it, the root cause you identified, the correction you applied, and the validation method you used to confirm the correction was effective. The story comes from professional work, not a personal project.

10. Understand the Copernicus Programme Architecture — the Six Sentinel Families, the Contributing Missions, and What the Data Access Model Means for EO Businesses 3–4 h

Skill: The dominant global EO data infrastructure as institutional context for the industry

Objective: The Copernicus programme is the world's largest EO data provider and shapes the economics of commercial EO more than any other single factor. Free, open access to Sentinel-1 (C-band SAR), Sentinel-2 (multispectral), Sentinel-3 (ocean and land colour, altimetry, surface temperature), Sentinel-5P (atmospheric chemistry), Sentinel-6 (altimetry), and the forthcoming expansion missions (CO2M, CHIME, CIMR, LSTM, Rose-L, Harmony) establishes a free baseline against which commercial data must compete on resolution, revisit, or timeliness. Engineers who understand what Copernicus provides and what it does not — and what commercial operators therefore have space to compete in — are demonstrating the institutional awareness that makes them useful in programme strategy discussions from their first week.

Done Signal: For each of the six Sentinel families, you can state: the primary instrument type, the primary application domain, the key spatial and temporal resolution parameters, and one specific limitation that a commercial operator might exploit. You can explain what the Copernicus Data Space Ecosystem provides programmatic access to. You can describe what one of the four forthcoming Sentinel Expansion missions (CO2M, CHIME, CIMR, LSTM) will add that the current Sentinel fleet does not provide.

11. Build Working Knowledge of Machine Learning for EO — Specifically What Makes Satellite Imagery Different from Natural Image Datasets 5–7 h

Skill: Applied ML with domain-specific rigour, not domain-agnostic pattern matching

Objective: Applied ML for EO is a large, active, and commercially important field — but one where naive application of standard computer vision produces unreliable results. Satellite imagery is acquired under varying illumination angles and atmospheric conditions that create distribution shift between training and deployment. It is often multispectral or hyperspectral, with bands beyond the RGB of natural image datasets. Temporal change is often the signal of interest rather than static scene content. Ground truth is spatially sparse and geographically biased toward accessible, well-documented regions. Engineers who design ML pipelines without accounting for these properties produce models that fail in deployment over new geographies, new seasons, or new sensor configurations in ways that are difficult to diagnose without domain understanding.

Done Signal: You can describe three specific ways in which satellite imagery datasets differ from ImageNet-style natural image datasets, and for each, describe a specific ML pipeline design decision that accounts for it. You have trained a classification or segmentation model on a real EO dataset (BigEarthNet, EuroSAT, SpaceNet, or equivalent), evaluated it correctly (without data leakage across geographic tiles), and can describe the dominant failure modes of the model in out-of-distribution regions.

12. Understand Tasking and Scheduling for Commercial EO Constellations — the Operational Product, Not the Orbital Mechanics 3–4 h

Skill: The commercial and operational reality of EO as a service

Objective: Commercial EO operators manage a constrained resource allocation problem across hundreds of customer requests, with constraints including downlink station visibility, onboard memory capacity, solar power availability, thermal limits, cloud cover probability forecasts, and sensor slew rate. Engineers working at commercial operators — Planet, Maxar, ICEYE, BlackSky, Satellogic — are expected to understand the tasking problem at the level of the constraints that shape it: why a SAR constellation can commit to revisit with higher confidence than an optical constellation in high-latitude, high-cloud-cover regions; what the trade-off is between wide-area survey and long dwell tasking modes.

Done Signal: You can explain the primary constraints that limit EO satellite tasking capacity — orbital geometry, downlink station visibility, onboard memory and data rate, thermal management, and sensor slew rate — and describe how each constrains the number of tasks that can be executed per orbit. You can explain why SAR tasking can offer higher availability guarantees for specific targets than optical tasking in maritime or high-latitude applications. You can describe how a cloud cover probability forecast enters a tasking optimisation and what the risk management implications are for the customer SLA.

13. Prepare for a Live Coding Exercise Involving Real Raster EO Data — Georeferenced Processing, Cloud Masking, and Index Calculation 4–6 h

Skill: Producing working geospatial code under interview conditions

Objective: EO algorithm and data engineering companies use take-home or live coding exercises involving real raster data. The typical exercise — read a multi-band GeoTIFF, apply a cloud mask, calculate a spectral index, aggregate to a coarser resolution or time composite, and validate against a reference — is tractable for an engineer who has done it before. For an engineer unfamiliar with GDAL's data model, rasterio's windowed reading, or CRS handling, the exercise produces bugs that are difficult to diagnose under time pressure. The specific failure modes — treating reflectance-scaled integers as floats without dividing by the scale factor, mixing CRS between datasets without reprojecting, computing NDVI from non-atmospherically-corrected data without noting the implications — are all invisible if the code runs without raising an exception.

Done Signal: You can implement, in under 60 minutes, a Python function that reads a multi-band GeoTIFF (e.g., a Sentinel-2 L2A scene), applies a cloud mask from the SCL band, calculates NDVI for cloud-free pixels, reprojects to a target CRS if needed, and writes a valid GeoTIFF output with correct georeferencing. You know where the silent errors are — scale factors, no-data handling, CRS mismatches — and you check for them explicitly.

Senior Level *(Principal Engineer, Technical Lead, Fellow)*

All mid-level tasks still apply. Calibration knowledge must extend to end-to-end uncertainty budgets traceable to SI units. SAR knowledge must extend to implementation experience, not just processing architecture.

14. Prepare to Walk Through the Complete Development of an EO Instrument or Processing Chain — from Requirements Through On-Orbit Validation 4–6 h

Skill: End-to-end system ownership at the level a senior EO engineer must demonstrate

Objective: Senior EO engineers at instrument manufacturers, mission operators, and space agencies

4–6 h

are expected to own technical threads from instrument specification through calibration approach through on-orbit commissioning through long-term performance monitoring. For an optical instrument, this means being able to discuss the instrument trade study (aperture vs. integration time vs. GSD vs. swath), the pre-launch calibration approach (integrating sphere absolute radiance, flat-field, spectral calibration), the on-orbit calibration strategy (solar diffuser, lunar observations, vicarious calibration sites), and the approach to detecting and correcting post-launch performance changes. The interview at this level is a peer discussion, and its quality depends entirely on the richness of the real experience behind it.

Done Signal: For one EO instrument or data processing chain you personally owned — not contributed to, owned — you can discuss without notes: the driving operational requirements, the key instrument design trade-off, the pre-launch calibration approach, the hardest anomaly discovered during on-orbit commissioning, how it was diagnosed and corrected, and what the long-term calibration stability looks like over the mission lifetime to date.

4–5 h

15. Prepare a Technically Grounded Position on One Active Frontier in EO — Hyperspectral from Commercial Constellations, SAR-Optical Data Fusion, or Foundation Models for EO

Skill: Forward-looking technical awareness at senior engineering depth

Objective: The EO field in 2025–2026 has several technically active frontiers where the engineering challenges are genuine and the solutions are not settled. Hyperspectral from small satellite constellations (Pixxel, Wyvern, HyperSat, CHIME) raises questions of SNR at reduced aperture, smile and keystone correction, and spectral unmixing algorithms appropriate for sub-metre-pixel mixed scenes. SAR-optical fusion for change detection raises questions of what information each modality contributes independently and what fusion architecture preserves radiometric fidelity while exploiting complementarity. Foundation models for EO — large self-supervised models pre-trained on petabyte-scale satellite imagery (Clay, Prithvi, SatMAE, GFM) — raise questions of spectral and temporal generalisation and where these models fail. Senior engineers who can engage at technical depth on one of these frontiers — not just describe the technology but identify the specific open engineering problem — are demonstrating the forward-looking technical leadership senior roles require.

Done Signal: You have a prepared 3-minute technical discussion covering one specific engineering challenge on one frontier — named specifically: the SNR penalty of pushbroom hyperspectral instruments on 16U CubeSats and what detector and integration time strategies partially recover it; the coherent change detection algorithm design for multifrequency SAR fusion; or the spectral generalisation failure of a pre-trained EO foundation model when applied to a new sensor's band configuration. Your position is technically grounded and connects your engineering background to the problem rather than describing it from the outside.

4–5 h

16. Prepare to Lead a Technical Discussion on EO System Architecture Trade-Offs for a Specific Application Scenario

Skill: Architectural decision-making at senior engineering speed

Objective: Senior EO engineers make or advise on system architecture decisions — sensor modality, orbit, processing chain, delivery latency — for specific applications where the requirements are in tension. “For a maritime domain awareness mission requiring identification of vessels down to 10 m in length, globally, daily — what is your sensor modality trade, and what constellation architecture does it imply?” requires reasoning through: the spatial resolution required for 10 m vessel detection (roughly 1–2 m GSD), the all-weather requirement (favouring SAR), the revisit frequency (requiring many satellites or wide-area spotlight SAR coverage), and the downlink and processing latency requirement. The senior interview tests whether you can structure this reasoning in real time, arrive at a specific conclusion, and defend it against follow-up challenges.

Done Signal: You have worked through at least three EO system architecture problems in your domain — stating the application requirement, reasoning through the sensor modality trade, estimating the constellation size and coverage architecture needed, identifying the key technical risk, and describing what you would need to change if one requirement changed. You can do this conversationally, without a whiteboard, in under 8 minutes per problem.

4–5 h

17. Understand the EO Data Product Value Chain — from Level 0 to Analysis-Ready Data to Derived Intelligence Products

3–4 h

Skill: Strategic product architecture awareness at senior engineering depth

Objective: The commercial EO value chain is moving rapidly up-stack. Raw imagery is becoming commoditised as constellations proliferate. The value is increasingly in analysis-ready data (ARD) — radiometrically and geometrically corrected, cloud-masked, and tiled into consistent formats — and in derived products: crop type maps, vessel detections, building footprints, surface deformation maps, wildfire perimeters, flood extents. Senior engineers who understand where processing adds value and where it destroys it — and can design a data product architecture that serves both data-literate downstream users and non-technical end customers — are valuable across the industry.

Done Signal: You can describe the processing level progression from Level 0 (raw instrument data) through Level 1 (calibrated radiance or top-of-atmosphere reflectance) through Level 2 (surface reflectance or equivalent geophysical variable) through Level 3 (composited or aggregated products) through Level 4 (derived thematic products). For each level transition, you can state what processing is applied, what quality information is added or lost, and what user capability the transition enables. You can name one specific Level 3 or Level 4 product and describe its product specification.

18. Build Quantitative Estimation Fluency for EO System Performance — Without Running a Simulation

4–5 h

Skill: Senior-speed system-level reasoning about resolution, SNR, coverage, and revisit

Objective: Senior EO engineers are called on to sanity-check instrument specifications, estimate whether a proposed constellation can meet a coverage requirement, and respond credibly to “is this achievable?” in programme reviews without access to a simulation tool. The estimation problems are tractable from first principles: computing achievable SNR from aperture and integration time, estimating swath width from field of view and orbit altitude, computing daily global coverage capacity from swath and orbital parameters, estimating InSAR deformation measurement precision from radar wavelength and baseline geometry, estimating constellation size for a given revisit requirement at a given latitude.

Done Signal: You have worked through five system performance estimation problems spanning at least two of: optical SNR from instrument parameters, GSD from aperture and orbit altitude, daily coverage from swath and orbital parameters, InSAR deformation precision from wavelength and coherence, or constellation size for a given revisit requirement at a given latitude. You state your assumptions before your numbers, arrive at order-of-magnitude answers, and can identify the most load-bearing assumption in each.

All Levels

19. Prepare a Specific Answer to “Why EO — and Why This Specific Modality or Application Area?” That Engages with the Field’s Current Technical Moment

1–2 h

Skill: Technically specific motivation that distinguishes field-committed candidates

1–2 h

Objective: EO is not a generic aerospace or data science role. Employers — particularly specialist instrument manufacturers, mission operators, and precision analytics companies — are looking for engineers who have engaged with a specific aspect of EO deeply enough to have a technical opinion about it: why the transition from 10 m to 3 m multispectral resolution changes the set of detectable agricultural stress signals; what the antenna aperture limitation means for commercial SAR achieving sub-metre resolution from a 150 kg satellite; why the forthcoming Copernicus CHIME hyperspectral mission will change the economics of the commercial hyperspectral market. The expected answer in 2025–2026 engages with one of: the hyperspectral commercial market opening, the SAR constellation proliferation, the EO foundation model transition, the precision agriculture and food security application, or the defence domain awareness requirement.

Done Signal: You have a 90-second answer that names your target EO modality or application domain, identifies one specific active engineering challenge in it, explains what makes that challenge technically hard (not just commercially important), and connects your background to contributing to its solution. The answer cannot be given unchanged to a candidate for a generic remote sensing or geospatial analytics role.

1–2 h

20. Map Your Technical Background to a Specific EO Employer Type, Modality, and Role — and Verify Security Clearance Eligibility Where Relevant

Skill: Precise self-positioning in an employer landscape that spans new-space startups to government agencies

Objective: The EO industry is extraordinarily varied. An algorithm engineer at Capella Space works in Python and C++ on a commercial SAR product in a startup environment, makes architectural decisions quickly, and ships code to production within weeks. A payload engineer at Airbus Defence & Space works in a structured programme environment with formal design reviews and long programme timescales. An imagery scientist at Planet works with petabyte-scale multispectral data and ML in a cloud-native data engineering environment. A research engineer at DLR or ESA works on exploratory algorithm development with longer research horizons and publication expectations. A candidate who cannot characterise which of these environments they are suited for is asking the recruiter to do analytical work they should be doing themselves.

Done Signal: You can state in under 60 seconds: the EO employer type you are targeting (new-space operator, legacy prime, instrument manufacturer, space agency, data analytics company, or government laboratory), the specific sensor modality most relevant to your background (optical, SAR, hyperspectral, thermal, or multi-modal), the programming languages and tools you work in fluently, your security clearance status or eligibility in your target country, and two specific technical contributions your background makes to the target role.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	Process Sentinel-2 scene to surface reflectance — programmatically	All	6–10 h
2	Electromagnetic spectrum as EO constraint — all regions, all physics	All	3–4 h
3	Image quality metrics — GSD, MTF, SNR, NEdL — instrument parameters	All	3–4 h
4	STAR stories — data quality, cross-discipline, self-correction	All	2–3 h
5	Georeferencing and geolocation accuracy — RPC model, terrain displacement	All	3–4 h
6	Hands-on SAR data — Sentinel-1 GRD, speckle filtering, backscatter	All	5–8 h
7	Radiometric link budget — from solar irradiance to detector SNR	Mid+	4–6 h
8	SAR processing depth — range-Doppler, InSAR coherence, resolution limits	Mid+	6–8 h
9	Calibration anomaly story — diagnosis, correction, validation	Mid+	2 h
10	Copernicus programme architecture — six Sentinel families, expansion	Mid+	3–4 h
11	ML for EO — domain-specific rigour, not domain-agnostic pattern matching	Mid+	5–7 h
12	Tasking and scheduling — commercial EO as constrained resource allocation	Mid+	3–4 h
13	Live coding readiness — raster processing, cloud masking, index calc	Mid+	4–6 h
14	Complete instrument or processing chain — requirements through on-orbit	Senior	4–6 h
15	One active EO frontier — technical position on a specific open problem	Senior	4–5 h
16	EO system architecture trade-off discussion — conversational	Senior	4–5 h
17	EO data product value chain — Level 0 to derived intelligence products	Senior	3–4 h
18	System performance estimation — five problems, stated out loud	Senior	4–5 h
19	“Why EO — why this modality or application area?” — technically specific	All	1–2 h
20	Employer type / modality / clearance self-positioning	All	1–2 h

THE STRUCTURAL REALITY

Earth Observation is a field defined by the gap between what the physics allows and what the application requires — and by the calibration, processing, and algorithm engineering that bridges it. A satellite detector is measuring photons reflected or emitted from the Earth's surface after passing through an atmosphere that absorbs, scatters, and refracts them unpredictably. The signal is real. The measurement is uncertain. The useful information is the difference between two uncertain measurements separated by days, months, or years.

The engineers who succeed in EO are those who take this uncertainty seriously — who understand what their calibration chain is actually guaranteeing, what atmospheric correction is actually removing (and what it is not), and what their machine learning model is actually detecting (and under what conditions it will fail). The quantitative value of EO data is entirely dependent on the rigour of the measurement chain, and the engineers who build and maintain that chain are doing science and engineering simultaneously, held jointly accountable by both disciplines.

The engineers who struggle are those who treat EO as a data source rather than a measurement system — who download images without understanding the radiometric scaling, apply atmospheric correction without knowing what the correction model assumes, train classification models without accounting for the geographic bias in their training data, and report accuracy metrics computed on data that leaked into training through spatial proximity to labelled pixels. These errors are invisible to a non-specialist. They are immediately visible to an experienced EO engineer.

BREAK INTO AEROSPACE
